

# machine learning

## 1- Variables and Functions

Understanding variables and functions is crucial for understanding how mathematical models work.

### Subtopics

- Variables
- Functions
- Domain and Range

## 2- Graphing and Plotting Points

Learning how to plot points on a graph and understand the relationship between variables.

### Subtopics

- Plotting Points
- Equation of a Line
- Function Graphs

## 3- Functions and Relations

Understanding what it means for one variable to be a function of another.

### Subtopics

- One-to-One Function
- Inverse Functions
- Composition of Functions

## 4- Polynomials and Expressions

Mastering polynomials and their operations with coefficients and exponents.

### Subtopics

- Polynomial Operations
- Degree of a Polynomial
- Factoring Polynomials

## 5- Matrices and Determinants

Applying matrices to represent and manipulate data, and calculating determinants to determine properties of matrices.

### Subtopics

- Matrix Multiplication
- Determinant Calculation
- Cofactor Expansion

## 6- Statistics and Probability

Understanding basic statistical measures and probability distributions.

### Subtopics

- Mean and Median
- Mode
- Standard Deviation
- Probability Basics

## 7- Data Analysis

Analyzing data sets to draw conclusions about population characteristics.

### Subtopics

- Descriptive Statistics
- Correlation and Regression
- Statistical Inference